Jacob Wilson

Biloxi, MS · (228) 302-6888 · jbw622@msstate.edu · linkedin.com/in/jbw622 · github.com/Aebyssian

Education

Ph.D., Computer Science, *University of Southern Mississippi* Fall 2026–Present

- First-year doctoral student

Bachelor of Science, Computer Science, *Mississippi State University* May 2026

- Mathematics minor · Shackouls Honors College · GPA: 3.47 overall / 3.72 major

Research

Quasi-Novelty / Reconciliation 2026–Present

Continual reinforcement learning, information rewards, and learning dynamics

- Undergraduate research adviser: Dr. Jingdao Chen, Mississippi State University.
- Built recurrent neural agents from random weights to study continual reinforcement learning. Trained agents with observation-derived rewards, without pretraining, demonstrations, or command-validity rewards; studied recurrent memory and learning from episode duration.
- Compared gradients with actual weight changes to identify how optimizer state suppressed encoder updates. Measured interactions between immediate and episode-end learning, and used entropy measurements to study policy collapse and responses to changing observations.
- Recorded 36 chains of dependent file reads across 26 distinct command sequences in one agent, verified against execution logs. The behavior was transient; eight completed runs from random weights had not reproduced it at the October 13 review.
- Recovered 24/24 correct responses from 0/24 on selected recorded inputs by reducing encoder offsets without retraining weights. Resuming RL from a collapsed checkpoint after one offset reset then produced 32 verified chains in the same agent; recovery was transient.

Working manuscript: *Quasi-Novelty Revisited.* October 2026.

Experience

Teaching Assistant, *University of Southern Mississippi* Fall 2026–Present

- Tutor students weekly and grade coursework for Computer Networks and Computer Systems.

Software Engineering Consultant, *HUMO, Starkville, MS* Feb–Oct 2023

- Developed motion-tracking software with faculty and graduate students for golf-swing analysis, applying sensor fusion and signal processing to real-time pose estimation from 9DOF and 6DOF IMUs.
- Built visualization tools in C#, Unity, and C; analyzed sensor data in Python; developed a Swift/iOS testing app and collaborated on a C++/Qt prototype.

Projects

Founder, Aebyss Technologies 2023–Present

- Built a Swift study-app prototype for an AI-based EdTech startup. Won second place and a $1,000 grant at the 2023 MSU Startup Summit; the startup is currently in Stage 3 of MSU's VentureCatalyst program.

Technical Skills

Languages:	Python, C#, C++, C, Swift, Rust, JavaScript
Tools:	PyTorch, Git, Unity, Qt, Xcode, React
Research:	Reinforcement learning, recurrent networks, linear algebra, numerical analysis